POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks: 100
Course: Discrete Structure | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Extended Euclidean algorithm with example. What is Inclusion-Exclusion Principle. 7

   b) Define Injective and Bijective Functions. Show that the function $f: R \dashrightarrow R$, given by $f(x) = 2x + 7$, is one-one and onto. 8

2. a) Let A = {a, b, c, d} and R = {(a, b), (b, c), (c, d), (b, a)}. Find the transitive closure of R using Warshall's algorithm. 8

   b) Define Equivalence and Partial order relation. Let R be the relation on the set of integers such that aRb if and only if **"a – b" is even**. Is R an equivalence relation? 7

3. a) Find the solution to the Recurrence Relation given by: 9
   $a_n = 6\,a_{n-1} - 11a_{n-2} + 6a_{n-3}$ where $a_0=2$, $a_1= 5$ and $a_2=15$.

   OR

   Explain Tower of Hanoi problem with its solution.

   b) What do you understand by Recurrence Relation? Explain the types of recurrence relation with examples. 6

4. a) Verify following argument is valid using the rules of inference: If you send me an e-mail message, then I will finish writing the program. If you do not send me an e-mail message, then I will go to sleep early. If I go to sleep early, then I will wake up feeling refreshed, leading to the conclusion that if I do not finish writing the program, then I will wake up feeling refreshed. 8

   b) Consider the following statement. Express below statement using quantifiers where domain consist of all birds. 7
      i. All humming birds are richly colored.
      ii. No large birds lives on honey.
      iii. Birds that do not live on honey are dull in color.
      iv. Humming birds are small.

   OR

   Explain different types of quantifiers used in predicate logic.

5. a) Explain Simple graph, Multigraph, Pseudo graph, with example. 5

   b) What is the Euler's formula for planar graph? Prove the case where adding an edge, vertices remain constant but Face increases. 5

   c) Differentiate between Hamilton graph and Euler's graph along with necessary diagrams. 5

6. a) What are the differences between DFA and N-DFA? Design DFA with $\sum = \{0, 1\}$ accepts even number of 0's and even number of 1's. 7

   b) Can we convert regular expression into finite automata? Explain with example. 8

7. Write short notes on: (**Any two**) 2×5
   a) Transport Network
   b) Pigeonhole principle
   c) Tautology and contradiction